

Power generation

ELECTRICITY IS PRODUCED IN DIFFERENT WAYS.

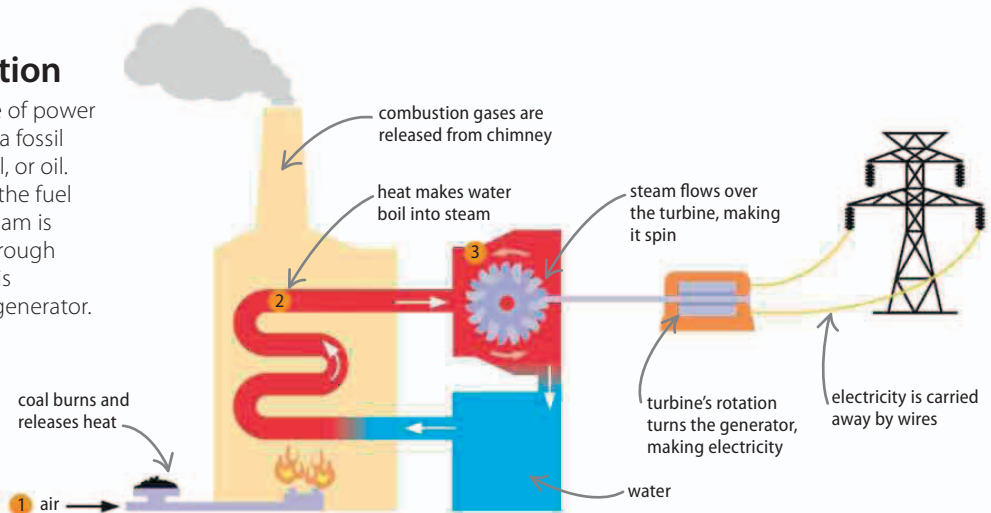
Electricity is generated on a large scale in power stations. They work in different ways, but they all harness a source of energy and use it to power giant electricity generators.

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Thermal power station

This is the most common type of power station. Its source of energy is a fossil fuel, generally natural gas, coal, or oil. The heat released by burning the fuel boils water into steam. The steam is forced under high pressure through the turbine, making it spin. This rotation is transmitted to the generator.



1. Heat from fuel

Solid fuels, like coal, are crushed into small particles, increasing its surface area so that it burns faster and hotter. The gases released by the combustion are released from a chimney.

2. Water into steam

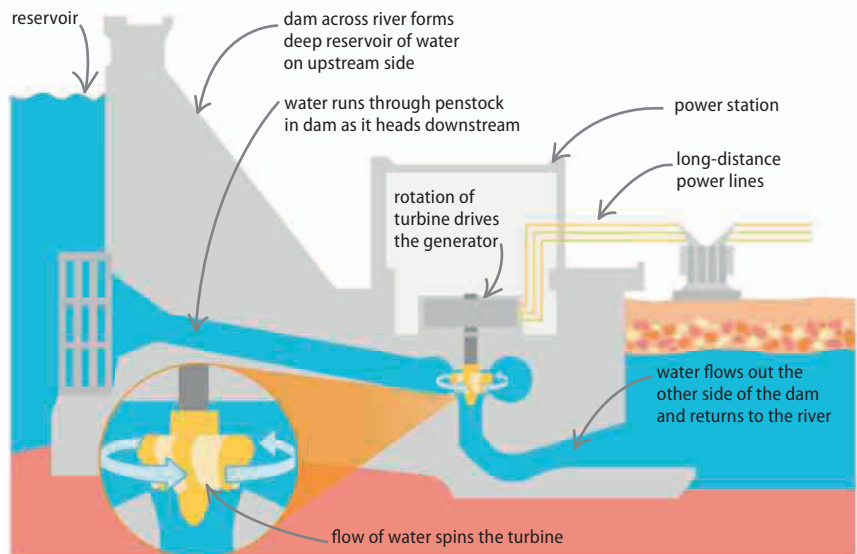
The water boils in the furnace, turns into steam, and passes over the propeller-like blades. The steam then condenses back into water to begin the process again.

3. Motion into electricity

The rotational motion of the turbine is passed to the generator, where a conductor is spun around in a magnetic field, inducing an electric current.

Hydroelectricity

In a hydroelectric power station, the energy of falling water is used to generate electricity. A dam built across a river builds up a large reservoir of water behind it. The water is released through a pipe, or penstock, to form a high-pressure flow that spins a turbine at the bottom. A water-driven turbine has cup-shaped blades, unlike the wing shapes on a gas or steam turbine.



▷ Power station

The turbine inside the dam is connected to a generator in a power station on the downstream side of the dam.